Parametric Integration

Exam code: 9MA0

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Written by: Paul
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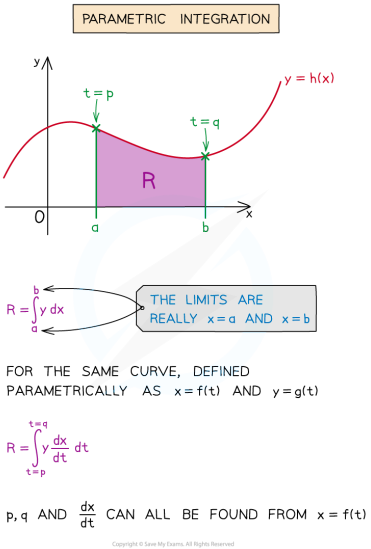
Parametric integration

How does integration work with parametric equations?

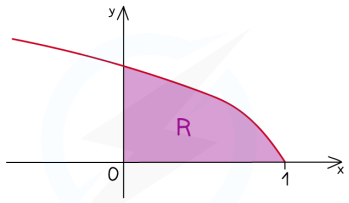
- Integration is used to find the area under a curve where the curve has been defined by parametric equations
  - **x = f(t)**
  - **y = g(t)**



Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

e.g. THE DIAGRAM BELOW SHOWS A SKETCH OF THE CURVE DEFINED PARAMETRICALLY BY

$$x = \cos 2\theta \qquad 0 < \theta < \pi$$
$$y = \sin\theta$$



FIND THE AREA MARKED R

STEP 1 WRITE DOWN THE INTEGRAL IN GENERAL TERMS OF x AND y

$$R = \int_0^1 y\,dx$$

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STEP 2 CHANGE BOTH LIMITS FROM x's TO THE PARAMETER

AT $x = 0$, $\cos 2\theta = 0$

$$2\theta = \frac{\pi}{2}$$
$$\theta = \frac{\pi}{4}$$

AT $x = 1$, $\cos 2\theta = 1$

$$2\theta = 0$$
$$\theta = 0$$

STEP 3 FIND $\frac{dx}{dt}$

$$\frac{dx}{dt} = -2\sin 2\theta$$

STEP 4 PUT THE INTEGRAL IN TERMS OF THE PARAMETER TOGETHER

CAREFUL WITH LIMITS, KEEP THEM IN THE RIGHT ORDER

$$R = \int_{\frac{\pi}{4}}^{0} -2\sin\theta \sin 2\theta \, d\theta \qquad (*)$$

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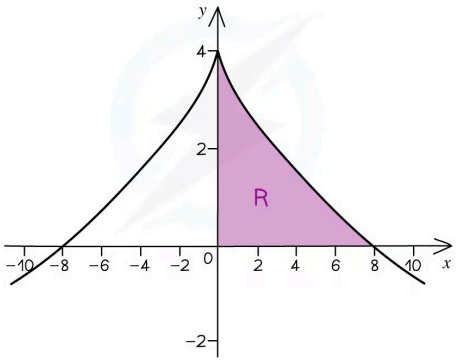
Worked Example

defined parametrically by

$$x = t^3 \text{ and } y = (4 - t^2)$$



Find the exact area of the region marked R.

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